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Genetic Algorithm-Based Acoustic Array Optimization for Estimating UAV DOA Using Beamforming

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Abstract: The localization of unmanned aerial vehicles is an important topic due to several threats near sensitive sites. Localization based on their sounds has been a particular point of interest in past studies for many years. It requires the use of a microphone array. The positioning of the various microphones making up an antenna defines the intrinsic directivity of the array. In this study, a genetic algorithm is used to determine the microphone positions that optimize directivity in a focus direction and for a frequency, by favoring the narrowness of the main lobe and the reduction of the secondary lobes. The optimization leads to several antennas with a 3D structure similar to that designed in a previous study. A method estimating the direction of arrival of a drone was also presented in that study making use of its acoustic signature to enhance the signal-to-noise ratio and thus improving the estimations. In this paper, an improvement to the method is proposed for tracking the drone's trajectory. Measurements were conducted to compare the drone locations given by the first designed antenna and the one optimized by the genetic algorithm. Performance on the direction of arrival found is characterized in terms of mean error, standard deviation and root mean square error relative to the GPS reference onboard the UAV. An experiment with the optimized antenna has also been conducted with the drone at a great distance to the antenna to characterize the maximal distance for possible estimations of the direction of arrival. Results show that the method used for the direction of arrival estimation can give a mean error below 10° in azimuth and 5° in elevation. The maximum distance between the antenna and the drone for which the method is able to give estimations is between 240 and 340 m.

Keywords: acoustic direction of arrival; beamforming; microphone array; genetic algorithm; time-frequency representation; experiments



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1. Introduction

Many opportunities are offered with the use of unmanned aerial vehicles (UAVs), like filming, photography, farm monitoring, search and rescue missions, cartography and many others. However some UAVs can become dangerous when used in sensitive areas or can be invasive in regards to privacy matters [1,2]. Ref. [1] reviews how UAVs are used both for good and malicious purposes. The review also describes the vulnerabilities of a drone and proposes several ways of classifying types of drones. Ref. [2] presents a review of different security threats in the particular application of UAV-aided networks. Dealing with these threats requires knowing the exact position of the UAV. Several methods exist but taking advantage of the sound produced by the UAV is of particular interest because of its unique and prevailing noise.

In [3], a review of different technologies is presented to detect, identify and locate drones and their controller. The technologies presented are radio frequency, radar, acoustic and electro-optical methods. It appears that radio frequency methods are not suitable to autonomous UAVs as they are limited by their size, which restricts the detection range. Radar also has difficulty distinguishing birds or pedestrians from drones. Electro-optical methods are limited by weather conditions or background temperature. Although acoustic methods are limited by disruptive noise and the sound power emitted by the drone, which must be high enough to enable localization, they have the advantage of being able to track autonomous drones, distinguish them from birds or pedestrians, be less sensitive to temperature and luminosity and be cost-effective. In the present work, it has, therefore, been decided to consider only approaches using acoustic signals collected by a microphone array even though merging the methods could make it possible to obtain competitive performance and accuracy for detection, identification and localization [4,5].

Lamotte et al. provide a comprehensive review of the methods used in the context of drone localization with acoustic sensors [6]. The methods are classified into three categories depending on whether they are based on time difference of arrival (TDOA), on a calculation of maximum energy, such as beamforming [7], or whether they offer high resolution like multiple signal classification (MUSIC) [8]. Some studies may also compare methods belonging to different categories. Ref. [9] compares the well-known TDOA method called generalized cross-correlation with phase transform weighting (GCC-PHAT) with conventional frequency domain beamforming on real experimental data obtained with a UAV. Conclusions of this study show that GCC-PHAT is faster but less accurate than beamforming. However, an improved version of GCC-PHAT based on a Bayesian filter and a Gaussian mixture model (GMM) is derived in [10], enabling good performance despite multipath effects and a low signal-to-noise ratio (SNR). As the traditional wideband MUSIC algorithm of the third category does not perform well under low SNR, a study proposed the MUSIC pseudo-spectral normalized mean processing technique to improve the performance at low SNR [11]. In the second category, several works use beamforming and specify the distance up to which it is possible to locate the drone. According to [12], the drone can be tracked using beamforming with a range of 160 m to 250 m. Another study using frequency domain beamforming is compared to the probabilistic parameter estimation technique called sparse Bayesian learning [13]. Results show a lower performance for beamforming with a localization range of 92 to 133 m compared to 104 to 152 m for the proposed method. Ref. [14] uses time beamforming while taking advantage of the drone's acoustic signature to enhance the performance of localization. Several studies showed that drones produce a harmonic signature with a fundamental frequency proportional to the rotation speed of the rotors [15–18]. Blanchard et al. used this harmonic signature to filter the signal using bandpass filters around the harmonics then performed time beamforming [14]. A similar approach has been presented in [19] using a time–frequency representation to select some specific spectral contents for the computation of energy after beamforming. This approach enables us to dispense of bandpass filters, which can be heavy in terms of computational cost when no phase shift is required. In this article, the time–frequency approach, by its ability to exploit the signature of UAVs, is retained with a complementary extension that takes into account the continuity of the drone's trajectory in order to improve the accuracy of the drone's tracking.

Acoustic localization relies on microphone arrays whose performance varies according to parameters such as the number of microphones and their position. An influential criterion is the spacing between microphones, which dictates the operating bandwidth. The arrangement of the microphones influences the sensitivity of the array, in particular the width of the main lobe in the beam pattern and the presence of sidelobes.

Some studies seek to compare existing arrays without questioning the microphone arrangement [20,21]. In these cases, the aim is to select the arrays that give the best results.

The antennas tested are often planar. The comparison in [20] is based on deconvolved spatial maps of sources from data coming from three classical arrays (circle, regular rectangular grid and spiral). The results show that the regular grid array has the lowest dynamic range in high frequency. The circular array gives the same performance as the spiral array, showing that with the use of a deconvolution algorithm, the spiral array can be replaced by a circular array. Another study consists of synthesizing spiral-like arrays by adjusting the dispersion of the distances between two microphones [21]. Antennas are selected after the evaluation of the main lobe width and the maximum sidelobe level (MSL).

Other work will seek to select from an antenna comprising many microphones, those which together will lead to a high-performance sub-antenna. Often these sub-arrays are defined as sparse arrays since their inter-sensor spacings exceed the Nyquist–Shannon criterion for spatial sampling $\frac{\lambda}{2}$, with λ the wave-number. In [22], sub-arrays of a uniform linear array are used to improve the accuracy of DOA estimation by relying on the Cramer Rao bounds. The criterion can also directly be a beampattern target. This is the case with the unified framework of sparse array, based on Lawson and the majorization–minimization algorithm, to find the microphone placement [23]. It is about beampattern synthesis for radar but could be applied to acoustic arrays.

Finally, one category of work aims to define an optimal antenna geometry in terms of certain criteria. These studies use different combinations of criteria and beampattern optimization algorithms. These algorithms are most often the particle swarm [24,25], simulated annealing [26], the minimax algorithm [27] or the genetic algorithm [28–31]. The criteria are linked to the characteristics of the beampattern (main lobe width, attenuation of the sidelobes [24,27,29], dispersion of the array directivity [30]) or to the performance obtained by a localization method (minimizing the mean square error and the standard deviation of TDOA [25], minimizing the difference between the beamformer’s output and a target [26], correlating statistical features of the microphone distribution with beamforming results [28], maximizing the ratio between the desired DOA and the sum of other directions using beamforming [31]).

In these studies, the antennas sought are mostly planar. Two studies looking at the optimization of a 3D antenna base their criteria on the beamforming result [24,28].

In the present study, the choice is made to optimize the placement of 10 microphones in a 3D structure divided into three axes [14] using the genetic algorithm and a criterion based on the main lobe width and the MSL. This structure was chosen for its logistical simplicity and the ability to track a drone with few microphones. The genetic algorithm was chosen for its ability to solve high-dimensional, non-linear optimization problems effectively. Its population-based approach ensures robust exploration of the solution space by maintaining diversity, which is especially crucial for navigating the 3D placement constraints in our application. While alternative methods discussed earlier are also applicable, the genetic algorithm’s proven success in solving similar sensor placement problems made it the most suitable choice for optimizing microphone placement. Moreover, genetic algorithms are particularly well-suited for incorporating multiple objective functions, which could be explored in future work to address additional factors such as localization errors or robustness to noise.

The contributions of this paper are as follows:

- A complementary extension of the time–frequency approach is presented, demonstrating improved tracking of the drone, taking into account changes in its position over time.
- A genetic algorithm is used to optimize a previously used 3D structured antenna. The results indicate that the optimization achieves similar performance in terms of DOA estimation but provides a clearer energy map, which could be useful for localizing multiple drones.

- Experiments were conducted to determine the maximum localization distance, which reached 340 m.

Section 2 presents a description of the acoustic signature of a drone. Section 3 presents the process followed to choose an optimized array with a genetic algorithm. Section 4 introduces the DOA estimation method with additional processing for drone tracking in the time–frequency representation method compared to reference [19]. Section 5 presents experimental results comparing the initial and optimized array with different flight scenarios. A scenario with a long distance between the drone and the array is also presented. Section 6 summarizes the contributions of this study.

2. Acoustic Signature of a Drone

As mentioned in the introduction, the signals emitted by drones have a harmonic structure. Figure 1 shows the spectrogram of an example of a signal picked up during the flight of a DJI Phantom 4 UAV, which starts on the ground, raises until around 2 m before returning to the ground. Harmonic frequencies are clearly present. The lowest, around 200 Hz, corresponds to the frequency at which the blades pass over the ground, which is referred to as the fundamental frequency. The other frequencies correspond to harmonics of this fundamental frequency, which itself is proportional to the speed of the rotors. As the drone has four rotors, at higher frequencies, around 800 Hz, several frequency traces can be observed.

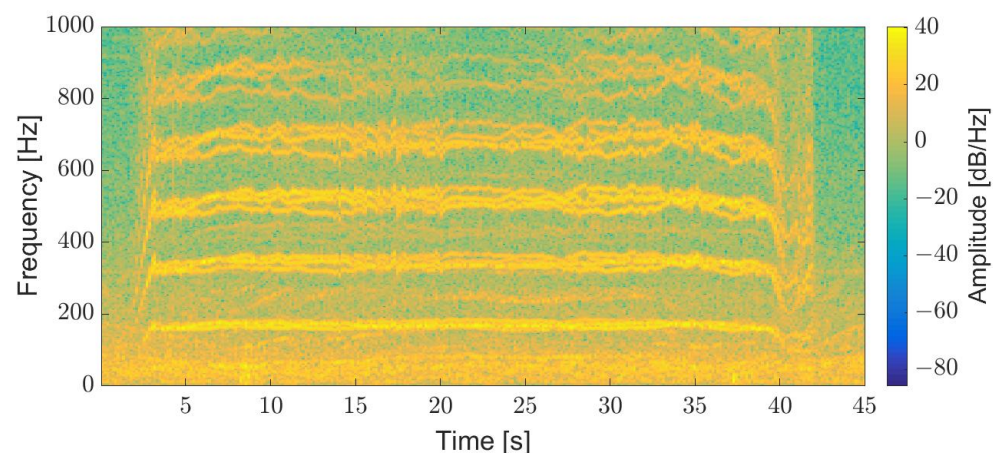


Figure 1. Spectrogram of the signal acquired by the first microphone of the array.

Some authors have tried to model the type of signals produced. Inspired by Bougaiov and Danik's work [32], a synthetic signal is proposed in [14] that contains two types of harmonics, weak and dominant harmonics. Weak harmonics are odd multiples of the rotor, and dominant harmonics are even with the first corresponding to the blade-passing frequency. Dominant harmonics amplitudes follow a logarithmic slope tuned with real measurements. In the work presented, the acoustic source constituted by the drone is considered to have several harmonics. The signal perceived by the antenna microphones will be assumed to have the same structure. The objective of the microphone antenna is, therefore, to highlight this harmonic structure as clearly as possible. This ability of the antenna depends on its geometry, hence the need to optimize the placement of the microphones.

3. Optimization of the Microphone Array

3.1. Initial Antenna Directivity

Our previous study [19] was conducted with a 10-microphone antenna with one microphone x_0 at the center and three microphones on each branch of a 3D axis

(see Figure 1 in [19]). The microphone has three different spacings for each branch in relation to microphone x_0 :

$$\begin{aligned} \|x_1\| &= \|x_4\| = \|x_7\| = l_1, \\ \|x_2\| &= \|x_5\| = \|x_8\| = l_2, \\ \|x_3\| &= \|x_6\| = \|x_9\| = l_3, \end{aligned} \quad (1)$$

with $l_1 = 5$ cm, $l_2 = 20$ cm, and $l_3 = 110$ cm. Microphones x_1 , x_2 and x_3 are on the first branch, microphones x_4 , x_5 and x_6 on the second, and microphones x_7 , x_8 and x_9 on the third. This arrangement gives a span of $L \simeq 155.6$ cm, which gives a low-frequency limit of $f_{min} = \frac{c}{L} = 220.5$ Hz for the array, and c denotes the speed of sound. The smallest spacing is l_1 , which gives a high-frequency limit of $f_{max} = \frac{c}{2l_1} = 3430$ Hz according to the Nyquist–Shannon sampling criterion. The directivity factor Q_d indicates how the antenna scans the space as a function of the frequency considered f and a focusing direction specified by its azimuth φ_r and elevation θ_r :

$$Q_d(f, \varphi_r, \theta_r) = \frac{|H(f, \varphi_r, \theta_r)|^2}{\frac{1}{4\pi} \int_0^{2\pi} \int_0^\pi |H(f, \varphi, \theta)|^2 \sin(\theta) d\theta d\varphi}, \quad (2)$$

with $H(f, \varphi, \theta)$ the directivity pattern [33]. Figure 2 shows an example of directivity factors for two different frequencies and focus directions. The characteristics of the main and side lobes are different, as shown in Table 1, which gives the main lobe widths in azimuth and elevation and the MSL for the two directivity factors. The lobe width increases as the frequency decreases and the number of sidelobes increases with the frequency. The aim is to propose an antenna whose microphone positions are optimized to promote the array's beam-pattern for the frequency and the focus direction considered. This optimization will be carried out using a genetic algorithm. When optimizing with a genetic algorithm, some constraints will remain the same as for the initial antenna, such as the number of microphones and the limit distances (minimum and maximum) between microphones on a branch.

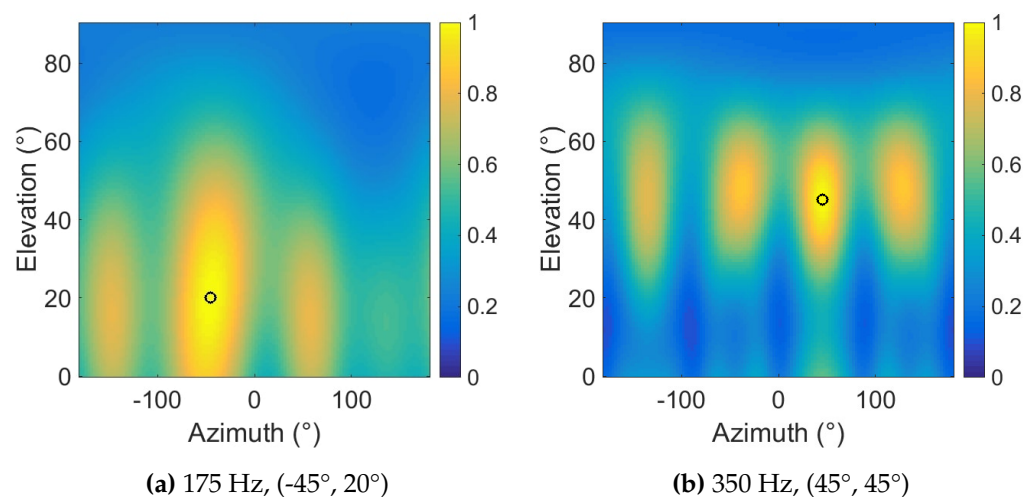


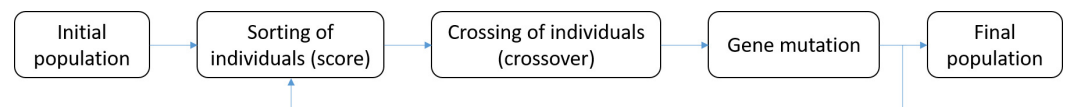
Figure 2. Directivity factor of the initial array for 175 Hz and $(-45^\circ, 20^\circ)$ focus direction (a) and for 350 Hz and $(45^\circ, 45^\circ)$ focus direction (b).

Table 1. Parameters (lobe widths and maximum sidelobe level) of the directivity factor for 175 Hz and $(-45^\circ, 20^\circ)$ focus direction and 350 Hz and $(45^\circ, 45^\circ)$ focus direction.

	175 Hz, $(-45^\circ, 20^\circ)$	350 Hz, $(45^\circ, 45^\circ)$
$lobe_width_{azi}$ ($^\circ$)	52.5	32.3
$lobe_width_{elev}$ ($^\circ$)	80.3	47.6
MSL (dB)	2	1.2

3.2. Genetic Algorithm

Genetic algorithms are inspired by nature and the evolution of a population throughout generations. The goal of the genetic algorithm is to find the microphone positions that minimize the main lobe and enhance the MSL to improve the directivity factor. The general process is described in Figure 3.

**Figure 3.** Genetic algorithm process.

The algorithm begins with an initial population. A population is constituted by individuals characterized by a genome, which is an ensemble of genes. In the case of a microphone array, a gene is the position of a microphone, the genome is the set of all positions, and an individual is an array, in other words, a set of positions given by the genes, for the 10 microphones. The initial population is generated randomly. In our case, for each array, one microphone is at the center, and the three microphones on each branch are free to be in any space, given that they respect the smallest spacing of l_1 and the largest spacing of 120 cm. The number of individuals in each population is 10.

The first process computed on the population is a sorting according to a criterion called the fitness function, which attributes a score to each array. The fitness function fit chosen here is based on two characteristics of the directivity factor Q_d , the main lobe azimuth width and the MSL such as:

$$fit = \frac{lobe_width_{azi}}{MSL}. \quad (3)$$

The directivity factor depends on the choice of a frequency and a focus direction. For the optimization process, the frequency is set at 350 Hz, which is compatible with a possible harmonic of the blade passing frequency of one of the UAV's rotors. The focus direction is arbitrarily fixed at $(45^\circ, 45^\circ)$.

After sorting the arrays, a crossing is performed called a crossover. The next generation is obtained by taking some genes from an array and others from a different array. Table 2 shows the combinations used to perform this crossover. The first array is kept, but other arrays are composed of the five first microphones of the father array and the four last microphones of the mother array. For example, the second child array is obtained by taking the five first microphones of the first array (the father) from the previous generation and the four last microphones of the second array (the mother) from the previous generation.

Table 2. Table of crossover to get from one generation to another.

Father array	1	1	1	1	1	2	2	2	3	3
Mother array		2	3	4	5	3	4	5	4	5
Child array	1	2	3	4	5	6	7	8	9	10

The last step of the algorithm is mutation. One microphone position is taken randomly and replaced by another random position. Once all these steps are completed, the process starts again from the sorting until a chosen number of generations is reached.

3.3. Optimization

The genetic algorithm is run with 100 and 200 generations for 350 Hz frequency and focus direction (45° , 45°), which corresponds to the direction in front of the array. The optimization is performed for one direction and one frequency since the fitness function depends on the directivity factor computed for one direction and one frequency. The frequency chosen corresponds to a harmonic of the average blade passing frequency of a drone during a flight. The direction of the directivity factor for the optimization is chosen arbitrarily for this study. Figure 4 shows the directivity at 350 Hz and (45° , 45°), for the initial array, the optimized array with 100 and 200 generations. The initial array highlights four lobes and thus a low MSL. The two optimizations show a much better MSL with a very low side-lobe level compared to the main lobe. The second optimization with 200 generations gives a better compromise than the two other arrays with a low MSL and a lower lobe width in azimuth. Figure 5 presents the arrays corresponding to the three configurations studied. The optimized array with 200 generations has fewer microphones close to the center in the azimuth plane and more microphones grouped together in the z axis compared to the initial array.

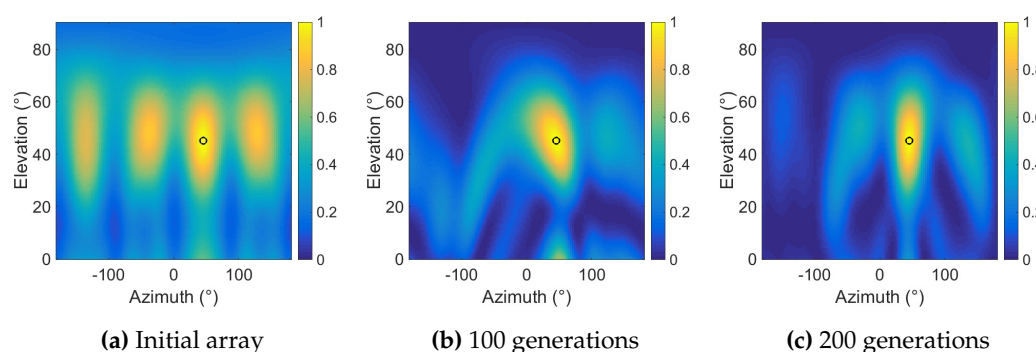


Figure 4. Directivity factor of (a) the initial array for 350 Hz frequency and direction (45° , 45°), the optimized array with (b) 100 generations, (c) 200 generations.

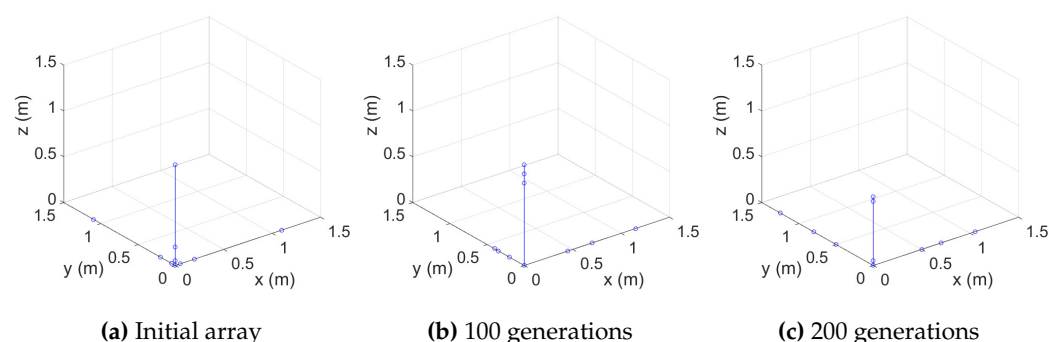


Figure 5. (a) Microphone arrangement for the initial array and the optimized arrays with (b) 100 generations and (c) 200 generations.

To compare the performance of the three arrays in all directions, maps of three directivity factor parameters can be used. For each direction, the directivity factor is computed, then the main lobe width in azimuth and elevation are measured, as well as the MSL. The maps are presented in Figure 6. For an elevation higher than 80° , the main lobe width in azimuth takes up the whole width for the three arrays, which shows that the arrays are not very sensitive when sources are above them. In other directions, the lobe width

increases with elevation for the three arrays. The optimized array with 200 generations still performs better in all the directions compared to the initial array with more widths close to 50° between 0 and 100° in azimuth and below -100° in azimuth. The first optimized array has more directions with widths in azimuth close to 50° but also has higher widths around 0° of azimuth than the other two arrays. For the elevation maps, the optimized arrays show a better sensitivity for directions above -100° in azimuth (in front of the array) compared to the initial array, but they have a lower sensitivity around -100° in azimuth (behind the array). The optimized array with 200 generations shows a better MSL compared to both other arrays with values close to 6 dB. The array which gives the best compromise for all the directions between the three arrays is the optimized array with 200 generations. This array is used in the rest of this study to carry out measurements. Table 3 provides the microphone positions of the chosen optimized array with 200 generations.

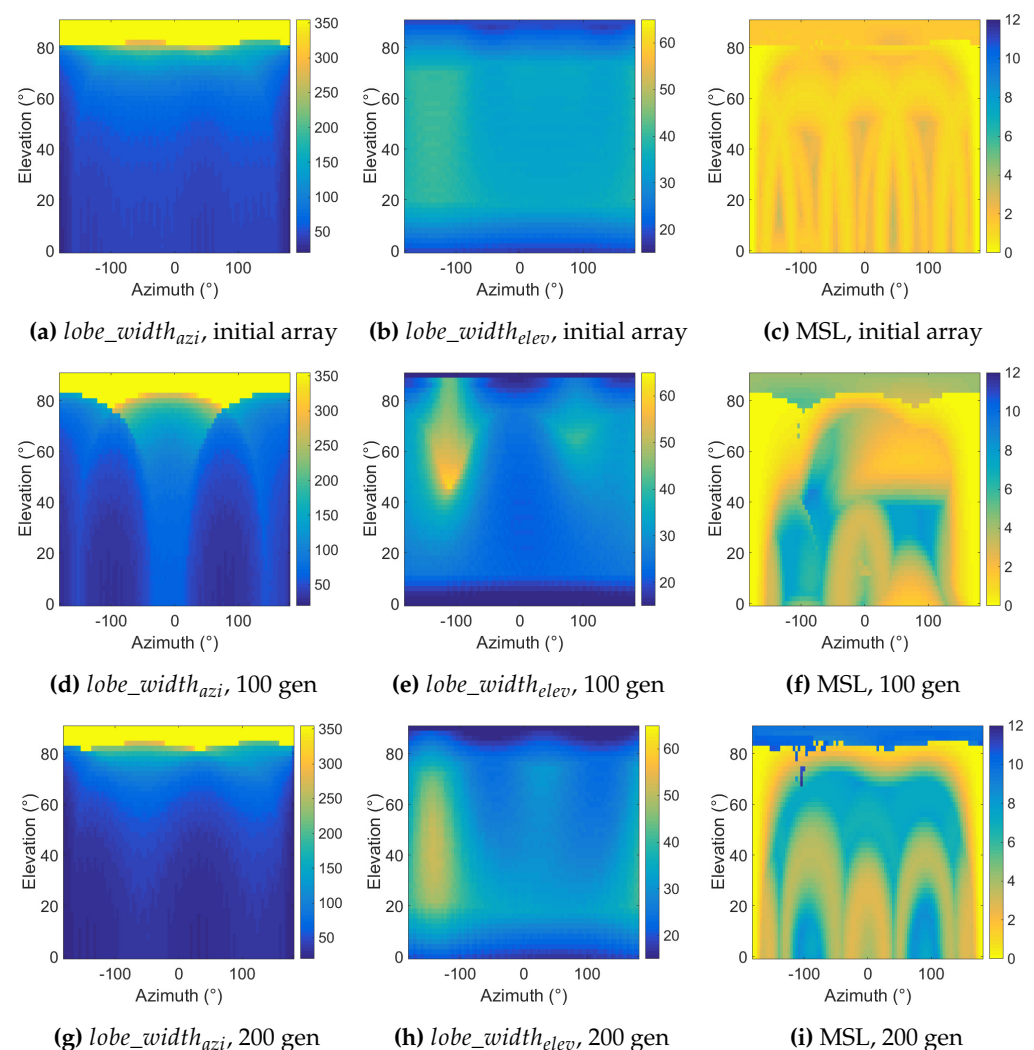


Figure 6. Maps highlighting the directivity factor parameters (lobe widths, MSL) for the initial array and optimized arrays with 100 and 200 generations.

Table 3. Microphone positions for the optimized array with 200 generations.

Optimized Array 200 Generations										
x (m)	0	0.5	0.7	1.05	0	0	0	0	0	0
y (m)	0	0	0	0	0.5	0.8	1.25	0	0	0
z (m)	0	0	0	0	0	0	0	0.05	0.7	0.75

4. DOA Estimation

4.1. Acoustic Signal Processing

4.1.1. Delay and Sum Beamforming

The time–frequency representation method (TFR) is based on the delay and sum beamforming method [7]. The method is parameterized by a source model (for the UAV) and a propagation model from the source to the microphones of the antenna. The source S is assimilated here to an acoustic monopole. The pressure produced by this point source at a distance r_{iS} between S and the i^{th} microphone, is expressed as a function of its acceleration flow rate $F_a(t)$ such that

$$y_i(t) = \frac{\rho_0 F_a(t - r_{iS}/c)}{4\pi r_{iS}}, \quad (4)$$

where ρ_0 denotes the density of the air. This model is based on the free-field propagation of spherical waves. In our case, the distance from the UAV to the antenna is greater than the acoustic wavelength, which justifies a far-field propagation model, i.e., plane waves. The signal received by the i^{th} microphone will, therefore, depend on the acceleration rate $F_a(t - \tau_i)$ delayed by a time depending on the direction of arrival of the source, i.e., its azimuth φ_r and elevation θ_r . This delay, linked to the i^{th} microphone, can be expressed in relation to a reference microphone on the antenna as follows

$$\tau_i(\varphi_r, \theta_r) = (\mathbf{x}_i - \mathbf{x}_0)^t \mathbf{n}_S(\varphi_r, \theta_r), \quad (5)$$

where $\mathbf{x}_i$ and $\mathbf{x}_0$ vectors, respectively, contain the spatial coordinates of the i^{th} and reference microphones, $\mathbf{n}_S$ is the 3D DOA vector, t is the transpose operator. Thus, beamforming consists here of focusing the waves in one direction by delaying each microphone signal according to the difference of distance to the reference microphone. The focused signal $y(t)$ for a DOA is achieved by aligning the signals, i.e., by compensating for the delays between each microphone and the reference due to the time of flight of the acoustic waves such that

$$y(t, \varphi_r, \theta_r) = \frac{1}{N} \sum_{i=0}^{N-1} y_i(t + \tau_i(\varphi_r, \theta_r)), \quad (6)$$

with N the number of microphones and $y_i(t)$ the signal acquired by the i^{th} microphone. By focusing the signals in several directions, a sampling of the space is performed. When the direction of the real source matches the focused direction, the focused signal should have the higher energy. The DOA can then be estimated by finding the direction that gives the maximum energy from the focused signals.

4.1.2. Time–Frequency Representation (TFR)

The TFR method uses beamforming, taking into account the knowledge of the drone's signature. The process, first proposed in [19], is described in Figure 7. For a given direction, described by the azimuth φ and the elevation θ , the signals are focused in that direction, then a short-time Fourier transform (STFT) is applied to the focused signal. Using the pitch-tracking algorithm defined below, given that the signal is harmonic, time–frequency bins corresponding to these harmonics can be selected. The selection of the time–frequency bins is conducted by defining the bandwidth surrounding the selected harmonics. Since the lines are more spread out as frequency increases, the bandwidths have been chosen here to vary with the frequency of the harmonic. The bandwidths Δf_k are defined such as:

$$\Delta f_k = \frac{f_k}{Q}, \quad (7)$$

with f_k the frequency of the k^{th} harmonic and Q , the quality factor chosen. Different bandwidths corresponding to different quality factors are illustrated in Figure 8. As with conventional beamforming, the energy of these time–frequency bins is calculated for one direction and then the process is repeated for several directions. The estimation of the DOA is given by the direction that maximizes the energy of the focused signal.

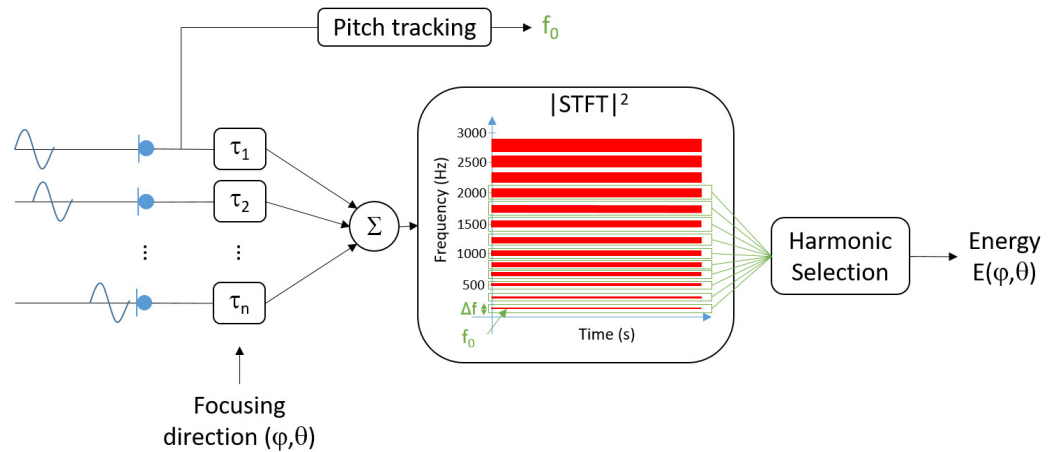


Figure 7. Synopsis of the time–frequency representation method.

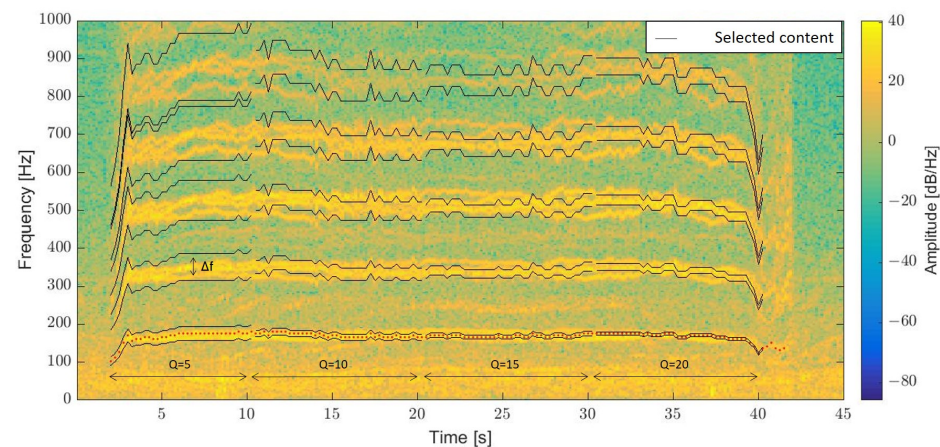


Figure 8. Spectrogram of the signal acquired by the first microphone with different bandwidths corresponding to different quality factors Q .

4.1.3. Spectral Harmonic Correlation

The pitch tracking algorithm used is the spectral harmonic correlation (SHC) described in [34]. This method calculates the correlation between a frequency and a chosen number of harmonics to detect if this frequency corresponds to the fundamental frequency of a harmonic series. The correlation is performed with windows around each considered frequency. The frequencies are sampled with a chosen range and the correlation is calculated according to:

$$SHC(t, f) = \sum_{f'=-\frac{L_w}{2}}^{\frac{L_w}{2}} \prod_{r=1}^{N_H+1} S(t, rf + f'), \quad (8)$$

with L_w the width of the windows, N_H the number of harmonics, and $S(t, f)$ the STFT of the signal. This method enables us to detect several candidates for the fundamental frequency of a signal. A simple criterion to discriminate the candidates is to take the frequency that maximizes the SHC. Readers can consult reference [19] for more details on the method and examples of results.

4.2. Automatic Choice of the Blade Passing Frequency

Experiments have been performed in order to characterize the performance of the TFR method as well as the pitch tracking algorithm. During this flight, the phantom IV performs one circle followed by another smaller one. Figure 9 shows the spectrogram of the signal recorded by the centre microphone of the array with estimations of the fundamental frequencies obtained with SHC using the maximum criterion. The frequencies fluctuate between 100 Hz and 170 Hz.

The azimuth and elevation estimates for conventional beamforming and the TFR method are presented in Figure 10. They are compared to the GPS embedded on the drone. The estimation methods were performed on 5000 points with a 50% overlap. A resolution of $(4^\circ, 2^\circ)$ was used for beamforming, corresponding to azimuth and elevation. Beamforming was performed with 4096 points and SHC with 8092 points. The TFR method was performed with 10 harmonics and $Q = 5$. In azimuth, conventional beamforming gives some good estimations at the beginning but shows a lot of estimations around -50° after 10 s. Similarly, in elevation, estimations are good for a few seconds at the beginning, then they fluctuate until the end. The TFR method enables us to have many more estimations close to the GPS both in azimuth and elevation. However, a lot of discontinuities are present for short times. In order to deal with these discontinuities, it is possible to use another criterion in the choice of the content selected in the TFR. Indeed, instead of taking the frequency that gives the maximum SHC, the idea is to take a frequency that gives a coherent estimation of the DOA. The frequency content selected in the TFR method associated with the candidate fundamental frequency must give an estimation of the DOA close to the previous DOA estimation. This condition will be called the continuity criterion since it takes into account the continuity in the trajectory of the drone. The continuity criterion for a direction $\Omega_t = (\varphi_t, \theta_t)$ at an instant t is defined as:

$$\begin{cases} |\varphi_t - \varphi_{t-1}| \leq 20^\circ, \\ |\theta_t - \theta_{t-1}| \leq 10^\circ. \end{cases} \quad (9)$$

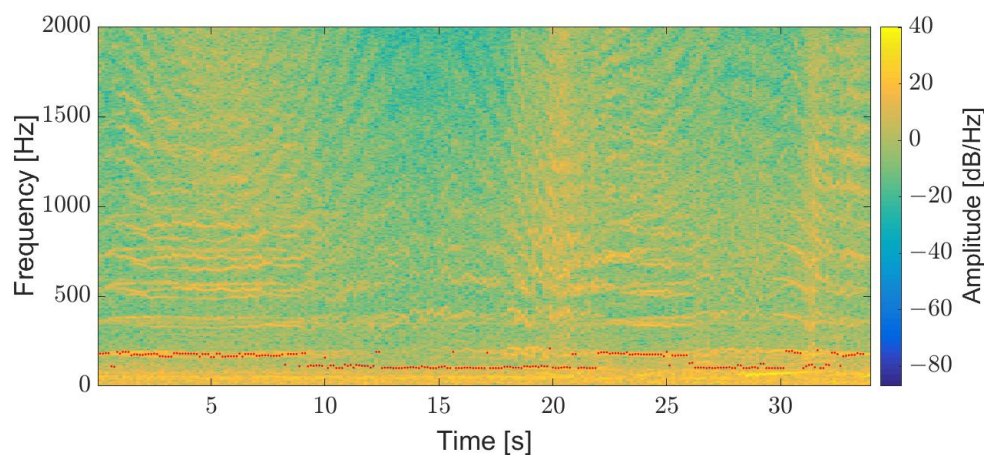


Figure 9. Spectrogram of the signal acquired by the first microphone of the array during the drone's circular trajectory using the maximum criterion in SHC. The red points are candidate frequencies whose harmonics are selected in the TFR.

Several candidates are tested until they reach the continuity criterion. The last line of Figure 10 presents estimations with the TFR method using this criterion. The performance of the tracking in both azimuth and elevation is much closer to the GPS than the other estimation methods. Table 4 also shows the improvement in performance due to the continuity criterion, measured using the root mean square error (RMSE) compared to the other two configurations. It is interesting to see what happens in an example at two successive instants. Figure 11 shows the energy maps as well as the SHC for $t = 9.125$ s

(first column) and $t = 9.25$ s (last 2 columns). For the first instant, the DOA estimation corresponds to the direction of the drone and the fundamental frequency chosen is around 175 Hz. The second instant gives a DOA estimation close to -90° in azimuth and 0° in elevation with a maximum in the SHC around 100 Hz, which is not realistic in terms of trajectory. The third column shows that taking the second candidate (180.7 Hz) in the SHC gives a coherent energy map and DOA estimation in relation to the previous instant.

Table 4. Root mean square error (RMSE) of estimation errors, in relation to the GPS onboard the UAV, in azimuth (φ) and elevation (θ) for the measurements with the circular trajectory.

	Azimuth φ ($^\circ$)	Elevation θ ($^\circ$)
Classical beamforming	111.2	35.9
TFR maximum detection	67.6	19.3
TFR continuity criterion	14.8	9.9

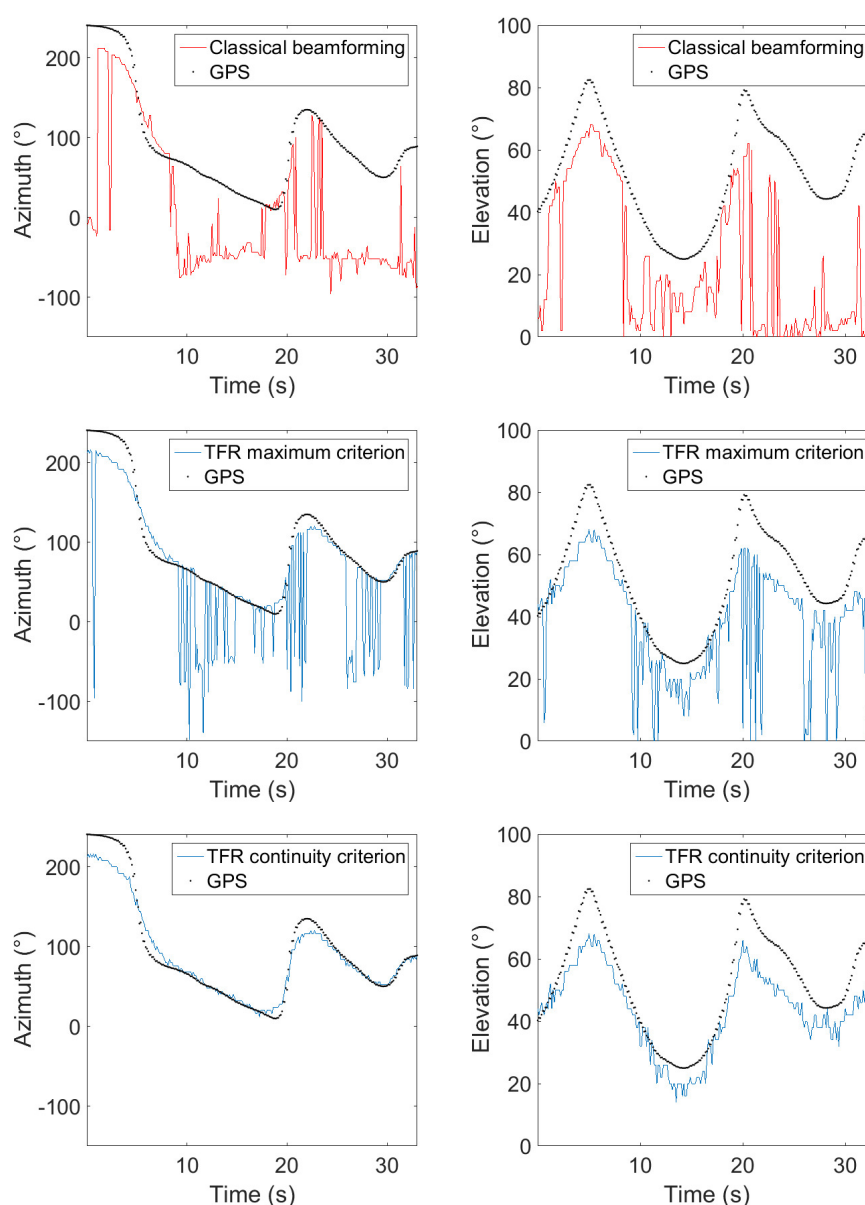


Figure 10. Azimuth and elevation vs. time for the circular trajectory with conventional beamforming and TFR method using two criteria for candidate selection in the SHC: maximum criterion (2nd line) and continuity criterion (last line).

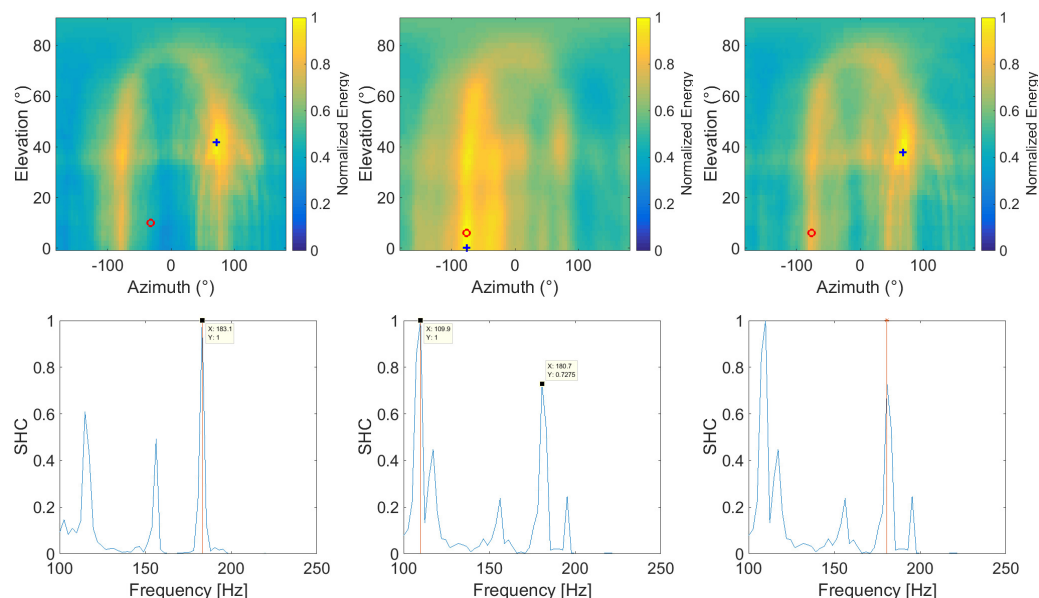


Figure 11. Energy maps for the TFR method at $t = 9.125$ s with the corresponding SHC (first column), at $t = 9.25$ s with the corresponding SHC using the maximum (second column) or the continuity criteria (third column). Crosses and circles correspond, respectively, to estimations obtained with the TFR and conventional beamforming.

5. Experimental Results

5.1. Comparison of the Initial and Optimized Antenna

Other experiments were performed in order to characterize the performance of the previously chosen optimized antenna compared to the initial antenna using the TFR method with the continuity criterion. The same trajectory is used and shown in Figure 12. The drone starts on the ground at around 40 m from the array, then ascends until 20 m, flies through a straight line toward the antenna then goes down close to the ground. It comes back to the start while ascending gradually then finally goes back to the initial point. This trajectory is performed with the two antennas, the drone flying in front of each antenna and behind each antenna. During the four measurements, sources of disturbance are present, such as the chirping of birds or the acceleration noise of car engines.

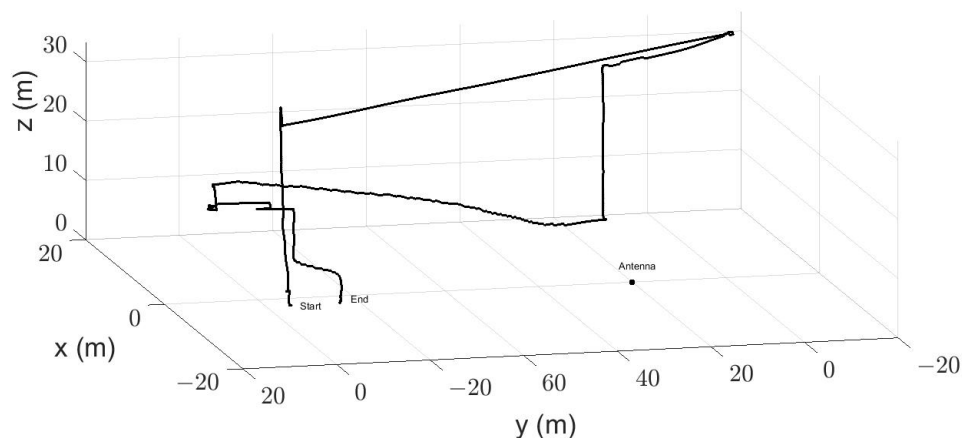


Figure 12. Trajectory performed by the drone for antenna comparisons.

Figure 13 presents the spectrograms of the four measurements (front and behind two antennas) with the corresponding candidates of fundamental frequencies estimated with SHC. Candidates are searched starting from 150 Hz for all cases except for the optimized

antenna in front, where too much noise is present and the search starts from 300 Hz. Indeed, the blade passing frequency around 175 Hz is masked by noise, so we choose to skip this harmonic and take the next harmonic, while still keeping 10 harmonics in the method calculation.

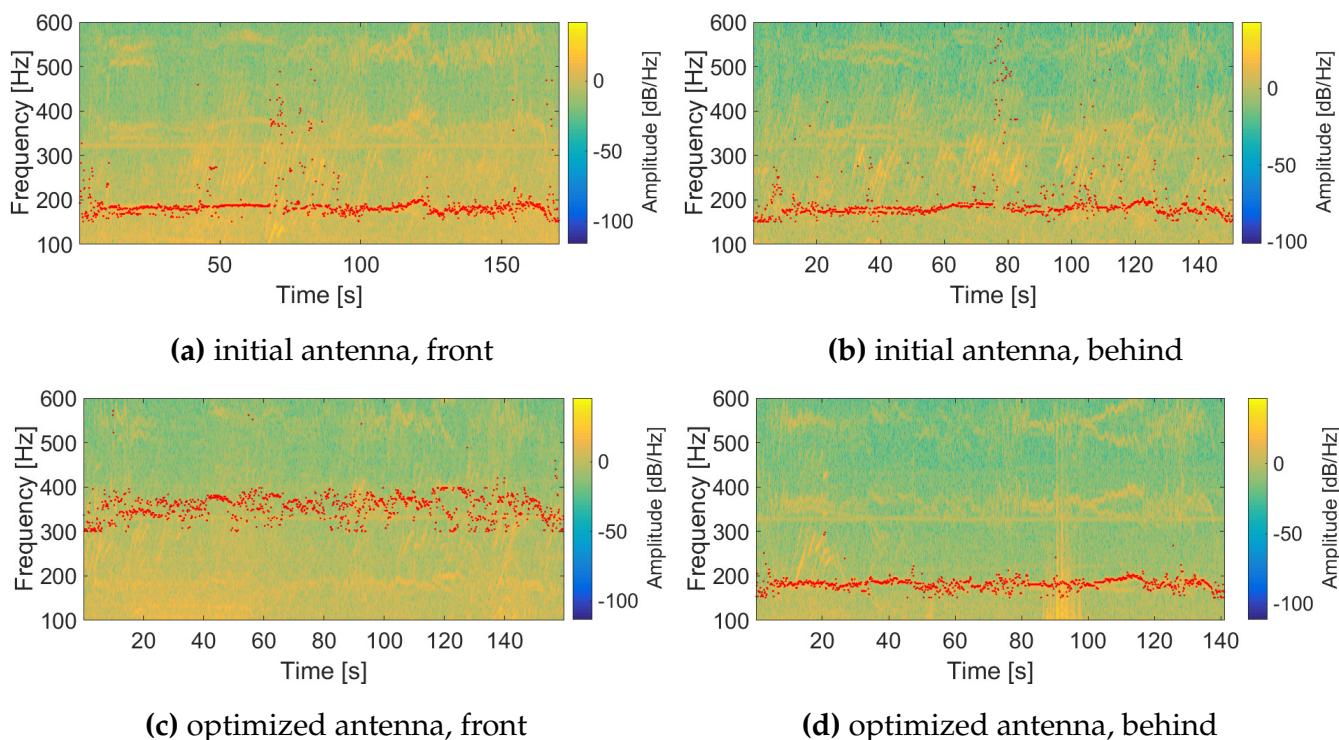


Figure 13. Spectrograms of the signals acquired by the first microphone of the array for the four measurements. The red dots are candidate frequencies whose harmonics are selected in the TFR.

DOA estimations for conventional beamforming for these measurements are presented in [35]. Results with the TFR method are presented in Figures 14 and 15 and show estimations very close to the GPS. Estimations do not vary too much between the four configurations. The optimized antenna shows some gaps between estimations in azimuth and the GPS at the beginning and at the end of the trajectory both in front and behind. Table 5 depicts the means, standard deviations and RMSE for each configuration. This table shows that in terms of DOA estimation in this scenario, the initial antenna performs better than the chosen optimized antenna but the performance results are close. DOA estimations with classical beamforming are not shown for the sake of clarity but the means, standard deviations and RMSE indicate lower performance compared to the TFR method.

The difference in performance between the initial and optimized antenna is more visible in Figure 16. Two instants are shown with energy maps given by conventional beamforming and the TFR method for the initial and optimized arrays with the drone in front of the antennas. These maps show that the TFR method enables us to obtain clearer energy maps with more localized energy. The optimized antenna shows a map that is much clearer and with a narrower main lobe, as compared to the initial antenna. The narrow main lobe can be highly beneficial in the case of multiple drone localization.

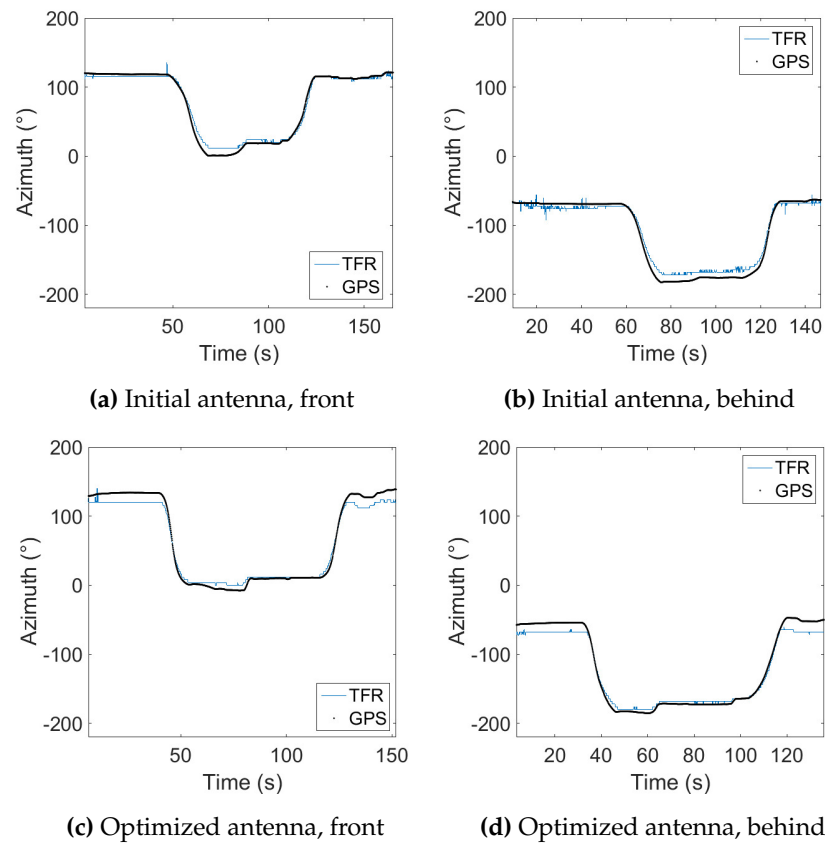


Figure 14. Azimuth vs. time for the TFR method in the four configurations of measurements.

Table 5. Mean (μ), standard deviation (σ) and root mean square error (RMSE) of estimation errors in azimuth (φ) and elevation (θ) for the measurements with the initial and the optimized antenna.

	Azimuth φ (°)			Elevation θ (°)		
	μ	σ	RMSE	μ	σ	RMSE
Initial antenna, front						
Classical beam,	79.8	76.3	110.4	15.3	15.0	21.4
TFR Method	4.2	3.7	5.6	2.7	2	3.4
Optimized antenna, front						
Classical beam,	86.5	58.6	104.5	22.0	15.7	27.1
TFR Method	8.1	6	9.7	3.8	2.3	4.4
Initial antenna, behind						
Classical beam,	85.6	61.7	105.5	16.4	14.3	21.8
TFR Method	6.7	3.8	7.7	2.7	2	3.3
Optimized antenna, behind						
Classical beam,	63.2	73.6	96.9	12.8	13.2	18.5
TFR Method	7.6	5.5	9.4	4.7	3	5.5

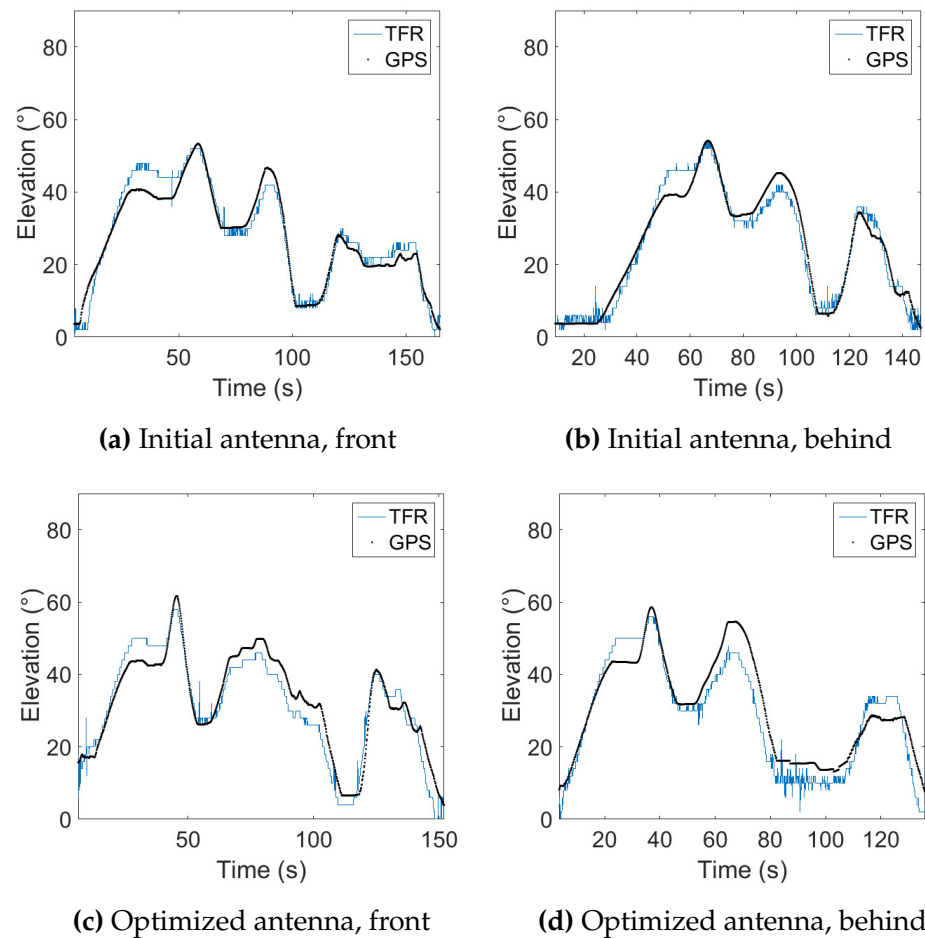


Figure 15. Elevation vs. time for the TFR method in the four configurations of measurements.

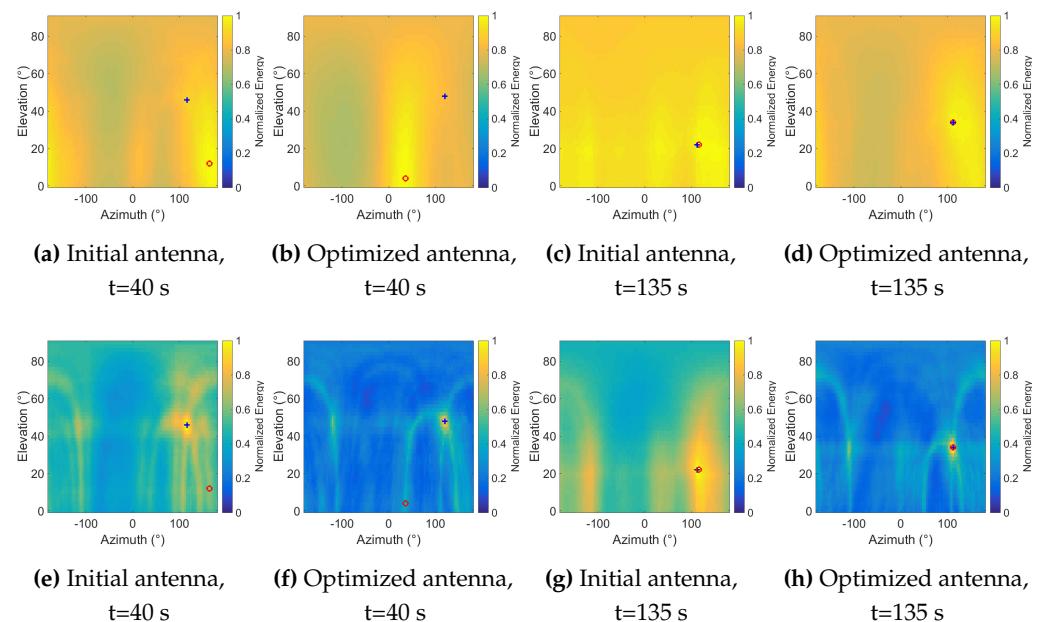


Figure 16. Examples of energy maps given by conventional beamforming (first line) and by the TFR method (second line) for two instants and two antenna configurations where the drone is in front of the antennas. Circles correspond to DOA estimations given by conventional beamforming and crosses those given by the TFR method.

5.2. Evaluation of the Maximum Distance for DOA Estimation

Another experiment was carried out to characterize the maximum distance for estimating the DOA. The trajectory performed by the drone is presented in Figure 17. The experiment was carried out in an open field close to a road. Disturbing sources are present during the measurement, such as car accelerations, bird chirpings and wind. Figure 18 shows the evolution of the distance between the drone and the antenna over time. The maximum distance traveled by the drone is 600 m. The spectrogram of the signal acquired by the first microphone is also shown in Figure 18. Many noises are included in this measurement below 100 Hz, and car accelerations are also present several times (30–40 s, 118–135 s, 161–176 s, 191–216 s, 290–300 s) (see the black circles in Figure 18a). It can be seen that the drone signal is no longer visible between 115 s and 216 s.

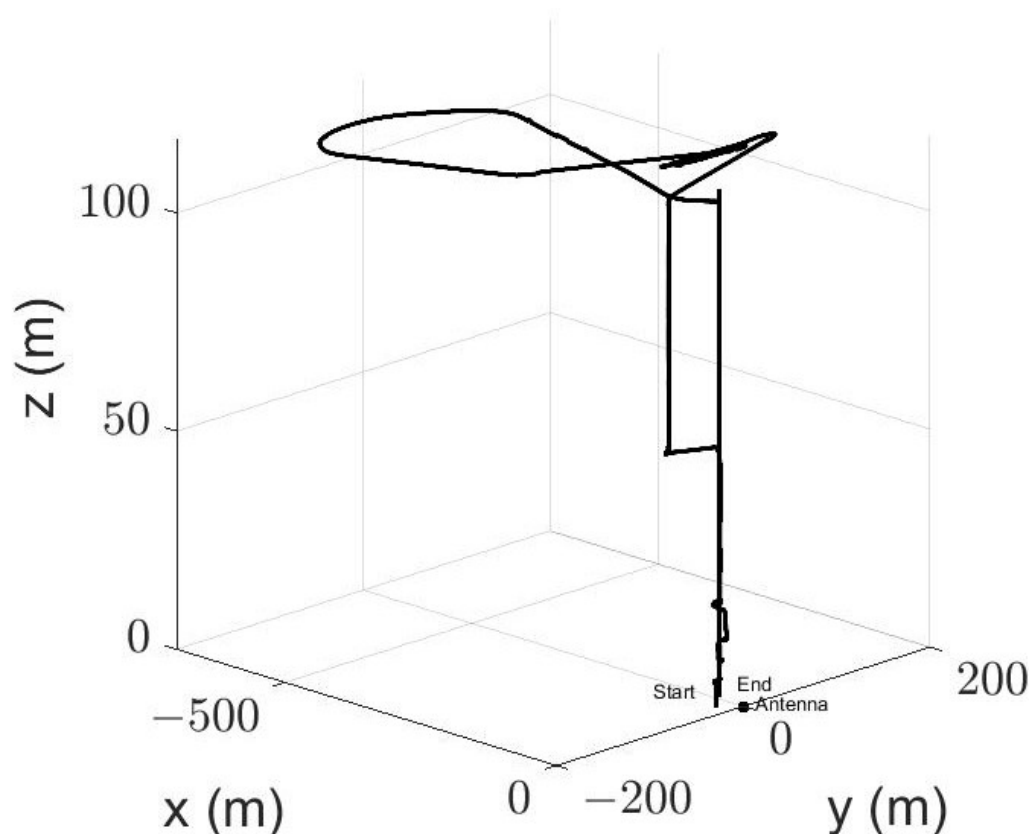


Figure 17. Trajectory performed by the drone for the long-distance measurement.

Figure 18 also illustrates that the DOA estimations given by the TFR method show a good tracking of the DOA until 119.5 s, which corresponds to a distance of 341 m. The azimuth matches the GPS between 142.8 s and 162.1 s, corresponding to distances between 594 m and 473 m. However, the elevation tracking is inaccurate during this period. Tracking resumes in both azimuth and elevation between 182.9 s and 189.3, corresponding to distances of 244.5 m and 242.9 m. Finally, the DOA estimations are correct from 223 s (177.3 m) until the end. It can be noted that the times when the drone is lost closely coincide with the presence of a car.

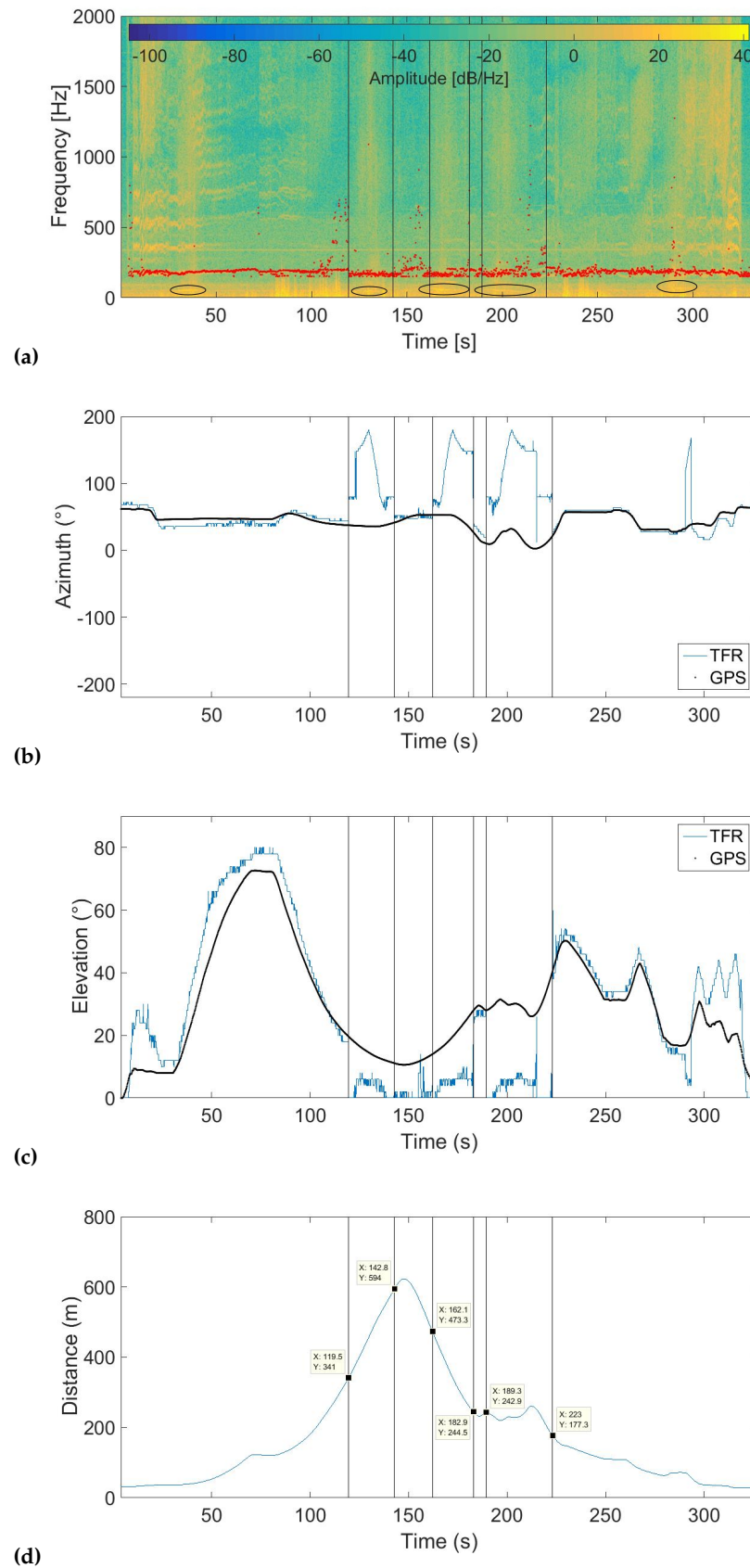


Figure 18. Spectrogram of the signal acquired by the first microphone of the optimized array during the long-distance measurement (a). Red points are candidate frequencies whose harmonics are selected in the TFR method. Areas representing car acceleration are circled. Azimuth (b) and elevation (c) vs. time for the TFR method. Evolution of the distance over time between the drone and the antenna according to GPS data embedded on the drone (d).

6. Conclusions

Acoustic sensors can be used to estimate the position or direction of arrival of a drone as a response to a privacy or security threat. Conventional beamforming is often used in source localization because of its robustness to noise, but this study shows that this method is very limited in terms of performance. However, taking into account the signature of the drone enables us to increase the signal-to-noise ratio and thus to enhance greatly the performance of the localization. A time–frequency representation method has been introduced in a recent study, and this paper proposes another strategy for obtaining better drone tracking by taking into account the previous points in the drone’s trajectory. The time–frequency approach is based on selecting the frequency content of the acoustic signal emitted by the drone, using a pitch-tracking algorithm. This frequency selection is crucial and dictates the performance of the localization. Further work is needed to evaluate the computational complexity of this approach to determine its suitability for real-time applications. Additionally, it would be interesting to test the method with different drones. Another key factor to consider in acoustic localization is the location of the microphones on the antenna. This study considered two antennas, one from the recent study already mentioned and another optimized using a genetic algorithm. The optimization and the choice of the antenna have been described, and experiments were conducted to compare the two antennas. In terms of localization performance, the two antennas are very close with the non-optimized performance being slightly better. This result may also be due to the fact that the optimization is carried out for a single frequency and a single direction. In any case, the methodology presented for optimizing the antenna is easy to apply and makes it possible to clarify the energy mapping while concentrating more energy in the target direction, which can be an advantage for locating multiple sources. Future work will involve exploiting optimization at several frequencies and focusing directions. Finally, experiments were conducted to evaluate the maximum distance of localization with the time–frequency representation method. Results show that the localization is possible up to 340 m when the drone was moving away and 240 m when it was coming back.

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